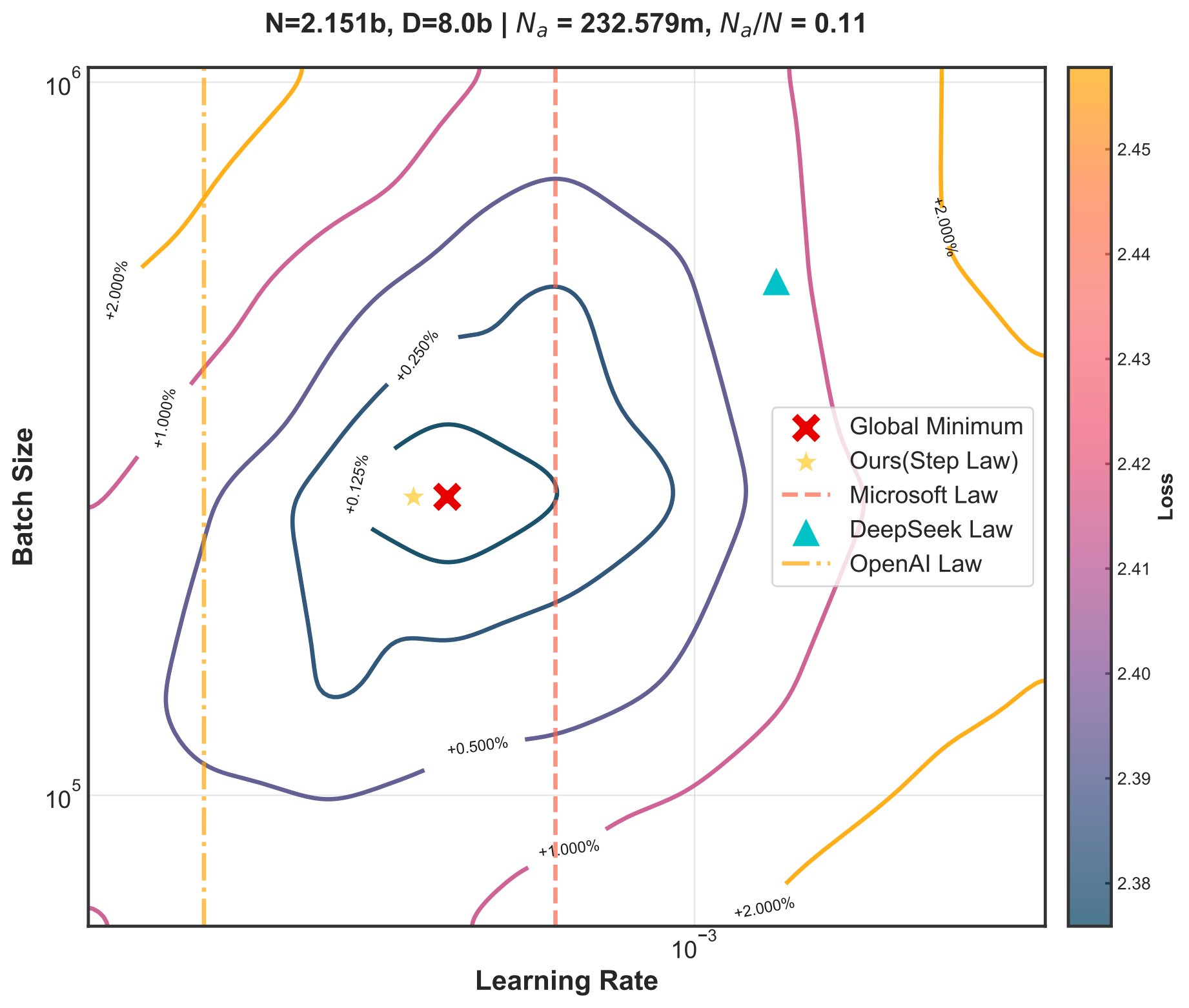
N=2.151b, D=8.0b | N_a = 232.579m, N_a/N = 0.11
Batch Size
Learning Rate
Loss
10^6
10^5
10^{-3}
2.45
2.44
2.43
2.42
2.41
2.40
2.39
2.38
+2.000%
+1.000%
+0.250%
+0.125%
+0.500%
+1.000%
+2.000%
+2.000%
Global Minimum
Ours(Step Law)
Microsoft Law
DeepSeek Law
OpenAI Law